Thesis Title

by

First_Name Last_Name

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in partial fulfillment of the

requirements for the degree of

Master of Science (M.Sc.)

in the program of

Applied Mathematics

Toronto, Ontario, Canada, 20XX

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Thesis Title

Master of Science (M.Sc.), 20XX

First_Name Last_Name

Applied Mathematics

Ryerson University

Abstract

Maximum of 150 words for a MSc thesis.

Acknowledgements

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Dedication

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References

[1] Surname, N. Imaginary publication. *Best Journal 0*, 1 (1900), 1–100.